

1. What it is

A single-tenant internal **finance & operations console** for the Waitara Holdings group (912 Holdings, Zone Fibre, Smart Zone Fibre, Global IT). It is a **vanilla PHP + MySQL/MariaDB** application, deeply integrated with **Zoho Books**, deployed on cPanel at nineonetwo.online/WORKINGCAPITAL. No framework and no build step — PHP serves pages and JSON; a hand-rolled JavaScript front-end drives the UI. **Zoho Books is the accounting source of truth**; this app is the front office.

2. The core workflow: Quote → Project → Invoice

The spine of the app is a job-costing pipeline:

1 · Quote — Built in the quote builder (line items: name, qty, rate, budgeted cost, VAT flag). Saved locally, then pushed to Zoho as an *estimate*. Each line carries a stable id (`lid`).

2 · Project — An approved quote is converted to a project (`is_project=1`). Technicians capture **actual costs** per line into the `project_costs` table, each tagged with a VAT mode (**No VAT** / **+16%** / **VAT-incl**). Costs attach to lines by `lid`, so reordering/deleting a line never misattributes cost.

3 · Invoice ("Bill client") — Admin bills the project: the app creates a Zoho *invoice* from the line items, pushes captured costs as Zoho *expenses* (tagged with the invoice number), and applies recorded client payments as Zoho *customer payments*. Guarded against duplicates (reconcile-by-reference + per-quote lock).

4 · Profit report — Per line: charged (ex-VAT) – actual cost = profit. **VAT is never part of profit**; it is reported separately (output VAT charged – input VAT paid = net VAT to remit).

3. Other modules

MODULE	WHAT IT DOES
Working capital / loans	Funds, loans to people/companies, repayments, outstanding balances.
Tasks	Task board with assignees, email notifications, calendar sync.
Reports	ETR (electronic tax register), invoice-status & quote-status reports (cached to <code>data/*.json</code>).
Email	Statements/reminders to clients via Zoho, an address book, draft management.
Ask AI	Chat answering questions about invoices, profit, and who owes money.
Client portals	Ben portal (separate <code>BENPORTAL</code> session, own login) and Audrey portal (public, token-gated) — limited external views.
Admin	User management (per-tab permissions), portal management, cost-booking config (Zoho accounts), WorkDrive backups.

4. Technical architecture

Front-end — `index.php` + `app.css` + `app.js`

- `index.php` (~85 KB): login page, app HTML shell, Ben portal, PWA manifest/service-worker, and a few inline endpoints (`?portal=ben`, `?invpdf` ...). Injects three PHP constants (`CFG`, `ME`, `ALLTABS`) then loads the cached `app.js`.
- `app.js` (~442 KB): the entire UI — per-tab `render()` functions, global state objects (QB, MQ, PROJ, LOANS, TASK, JC...), and ~134 `fetch()` calls. A hand-rolled SPA updating the DOM via `innerHTML` — no React/Vue.

Back-end — `api/*.php` (~71 endpoints)

Each endpoint: `session_start()` → `csrf_guard()` (Origin check) → auth check (`$_SESSION['auth']`) → `db()` → work → JSON. Errors pass through `err_ref()` / `api_fail()` , which log structured detail server-side and return a generic (`ref XXXXXX`) to the client.

HELPER	RESPONSIBILITY
<code>db.php</code>	PDO connection (from <code>config.php</code>).
<code>zoho.php</code>	Zoho OAuth + Books API; retry/backoff + concurrency-safe token refresh; VAT tax-id lookup.
<code>quote_pricing.php</code>	<code>quote_price()</code> — subtotal, per-line VAT, discount-before-tax, cost, profit.
<code>project_costs.php</code>	Cost table + Zoho expense sync + <code>quote_ensure_lids()</code> (line-id migration).
<code>errors.php</code> / <code>csrf.php</code>	Structured error logging (with SQLSTATE) / CSRF Origin guard.

Data

- **MySQL/MariaDB, 18 tables** (`quotes` , `project_costs` , `project_payments` , `project_assignees` , `fund_loans` , `tasks` , `app_users` ...) — auto-created/upgraded at runtime (self-healing DDL); canonical `sql/schema.sql` .
- `quotes.line_items` is a **JSON blob**; actual costs live relationally in `project_costs` , keyed to lines by `lid` .
- `data/` holds JSON caches (report results, the Zoho token, VAT tax-id cache) — web-denied via `.htaccess` .

Zoho Books integration (`zoho.php`)

A long-lived refresh token (in `config.php`) mints short-lived access tokens cached to `data/token.json` . REST is used for estimates, invoices, expenses, customer payments, contacts, taxes, and chart of accounts, plus WorkDrive for backups. Once data is pushed to Zoho it stays there even if deleted locally.

5. Authentication & permissions

- **Login**: a master `app_password` (in `config.php`) grants admin; per-user accounts live in `app_users` with hashed passwords. Sessions are hardened — SameSite=Lax, HttpOnly, Secure (via `.user.ini`).
- **Two tiers**: admin vs staff. Staff get cost-only views (prices/profit stripped server-side). Per-tab permissions (`costcap` , `projects` ...) and per-project assignees gate visibility.

6. Deployment

Git → GitHub (`twaitara/912-finance-console`) → cPanel pulls and runs `.cpanel.yml` , which **lints all PHP** and **runs the pricing tests** before copying files, then `cp` s root PHP, `app.css/js` , `api/` , and hardening `.htaccess` to the live path. `config.php` (secrets) is git-ignored — it lives only on the server and locally, never in the repo.

7. Quick file map

FILE	ROLE
<code>index.php</code>	Login, app shell, portals, inline endpoints, PWA
<code>app.js</code> / <code>app.css</code>	The entire front-end UI
<code>api/*.php</code>	~71 JSON endpoints (quotes, costs, invoicing, loans, tasks, reports...)
<code>zoho.php</code>	Zoho OAuth + Books API
<code>quote_pricing.php</code>	Quote / VAT / discount / profit math
<code>project_costs.php</code>	Cost capture + Zoho expense sync + line-id migration
<code>errors.php</code> / <code>csrf.php</code>	Error logging + CSRF guard
<code>db.php</code> / <code>config.php</code>	DB connection / secrets
<code>sql/schema.sql</code>	Canonical DB schema
<code>.cpanel.yml</code>	Deploy script (lint + test gate)